

Zorn's Lemma

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1 Notes

From https://www.acsu.buffalo.edu/~achirvas/Math461_Fall2018/conrad_zorn.pdf.

Definition: A partially ordered set is a nonempty set S with $A \subset S \times S$ such that $(s, s) \in A$ for all s , if $(s, s') \in A$ and $(s', s) \in A$, then $s = s'$, and if $(s, t) \in S$, $(t, w) \in A$, then $(s, w) \in A$.

Definition: A totally ordered set is a partially ordered set S with A such that for all $s, s' \in S$, $(s, s') \in A$ or $(s', s) \in A$.

Examples: Total ordering: $(\mathbb{Z}, \leq)$, $(\mathbb{R}, \leq)$. Partial ordering: $(P(\Omega), \subset)$, $(\mathbb{Z}^+, |)$.

Let X, Y be nonempty sets. Let $A \subset X$ be nonempty. Consider the set S of ordered pairs (A, f) , where $f : A \rightarrow Y$ is a function. Say $(A, f) \leq (B, g)$ if $A \subset B$ and $g|_A = f$. Then $(S, \leq)$ is a poset.

We can restrict posets to make smaller posets. Let $(S, \leq)$ be a poset and $T \subset S$. Then $(T, \leq)$ is a poset.

Let S be a poset and $T = \{s_1, s_2, \dots, s_n\}$ be a finite totally ordered subset of T . Then there exists an i for which $s_j \leq s_i$ for all $j \in [n]$.

Pick $s_1 \in T$. If there do not exist any $s_j \in T$, $j \neq i$ such that $s_1 \leq s_j$, then $s_j \leq s_1$ by total ordering, showing s_1 is maximal. Else, $s_1 \leq s_j$. We repeat this process inductively, so $s_{a_1} \leq s_{a_2} \leq \dots \leq s_{a_n}$ for some permutation (a_i) . By transitivity, $s_{a_i} \leq s_{a_n}$ and we have a maximal element.

Definition: An upper bound of $T \subset S$ is an element $s \in S$ such that $t \leq s$ for all $t \in T$. s is not necessarily in T , and could be a “bad” approximation.

Definition: A maximal element $m \in S$ is such that for all cases where m, s are comparable, $s \leq m$. Equivalently, the only s such that $m \leq s$ is $s = m$.

Zorn's Lemma: If every totally ordered subset of a poset S has an upper bound,

then S has a maximum.

Definition: A greatest element $m \in S$ is such that $s \leq m$ for all $s \in S$. m is a maximal element that can be compared to everything.

Definition: A least element $m \in S$ is such that $s \geq m$ for all $s \in S$. A least element is a minimal element that can be compared to everything in S .

The additive group $\mathbb{Q}$ has no maximal subgroups. Proof: Let $H < \mathbb{Q}$ be a proper subgroup. We will construct a group K such that $H < K < G$. Suppose FTSOC $[\mathbb{Q} : H]$ is finite, say of order n . Then $H = n\mathbb{Q} = \mathbb{Q}$, contradiction. Take any $r \in \mathbb{Q} - H$ and let $K = \langle r, H \rangle$. Note by the Second Isomorphism theorem,

$$K/H = (\langle r \rangle H)/H \cong \langle r \rangle / (H \cap \langle r \rangle).$$

Pick any $h \neq 0 \in H$. There exists a rational k that is an integer multiple of r and h , and in particular $k = mr$. Hence, $\langle k \rangle \subset \langle r \rangle \cap H$. We have $[K : H] \leq [\langle r \rangle : \langle mr \rangle] = m$. Since $[K : H] < \infty$ and $[\mathbb{Q} : H] = \infty$, $[\mathbb{Q} : K] = \infty$ and K is indeed a proper subgroup of $\mathbb{Q}$.

Question: is every subgroup of a finitely generated group also finitely generated?

Answer: no. Take $\langle y^n xy^{-n} \rangle$, $n \geq 1$, free on countably many generators.

Lemma: Let G be finitely generated. Then every subgroup $H < G$ is contained in a maximal subgroup.

Proof: Let $S = \{K < G : H < K\}$. S is a poset by inclusion. Consider a chain $K_1 \subset K_2 \subset K_3 \subset \dots$ and $L = \cup_{i=1}^{\infty} K_i$. L is closed under group multiplication: if $g, h \in L$, there exists m for which $g, h \in L_m$, so $gh \in L_m$. L is closed under inverses. Hence, L is indeed a group. L contains each K_i . It remains to show $L \in S$, or that $L \neq G$. Suppose FTSOC $L = G$ and that $G = \langle g_1, g_2, \dots, g_n \rangle$. For each $g_i \in G$, there exists m_i such that $g_i \in K_{m_i}$. Let $m = \max_i m_i$. Then $K_m = G$, contradiction. *Chain is not necessarily countable, but the argument still holds.